

|               |               |                  |
|---------------|---------------|------------------|
| Surname       | Centre Number | Candidate Number |
| First name(s) |               | 0                |



GCSE  
3445U20-1



MONDAY, 10 JUNE 2024 – MORNING

APPLIED SCIENCE (Double Award)  
UNIT 2: Space, Health and Life  
FOUNDATION TIER

1 hour 30 minutes

ADDITIONAL MATERIALS

- Separate Resource Folder
- Calculator
- Pencil
- Ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

| For Examiner's use only |          |              |              |
|-------------------------|----------|--------------|--------------|
|                         | Question | Maximum Mark | Mark Awarded |
| Section A               | 1.       | 8            |              |
|                         | 2.       | 5            |              |
|                         | 3.       | 6            |              |
|                         | 4.       | 10           |              |
|                         | 5.       | 5            |              |
|                         | 6.       | 3            |              |
|                         | 7.       | 4            |              |
|                         | 8.       | 9            |              |
| Section B               | 9.       | 6            |              |
|                         | 10.      | 19           |              |
|                         | Total    | 75           |              |

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 8(b) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

You will need to refer to the separate resource folder to answer questions 9 and 10.

A Periodic Table is printed on page 28.



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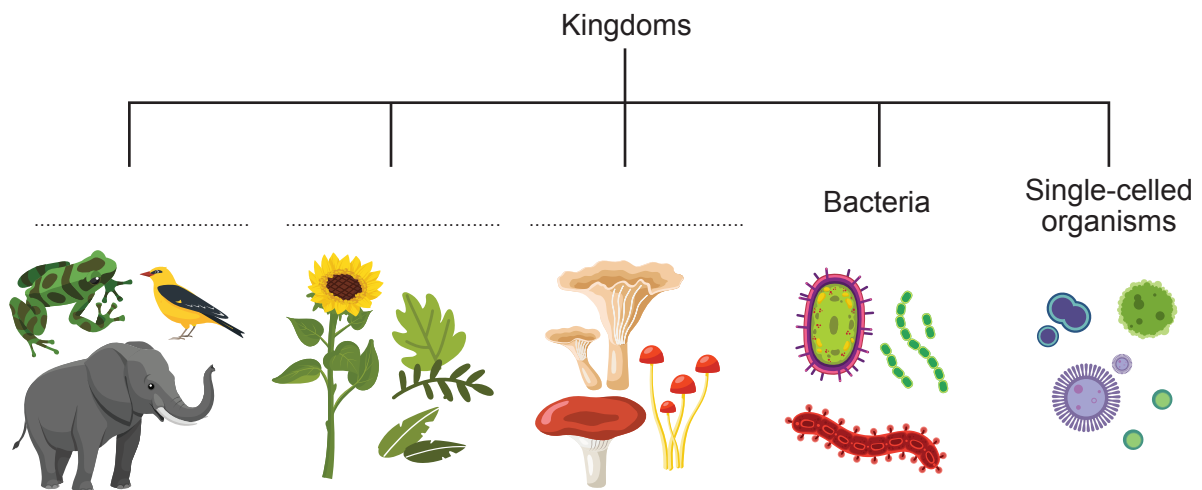
**Section A**Answer **all** questions.

1. Living things are classified using a system developed by Carl Linnaeus.

(a) The first division of living things is to place them into a kingdom. There are 5 kingdoms.

Complete the labelling of the kingdoms on the diagram below.

[3]



Examiner  
only

- (b) Linnaeus’ system further ranked living things to a species level. This is illustrated below. The levels are not in the correct order.

|         |       |       |        |       |         |        |
|---------|-------|-------|--------|-------|---------|--------|
| A       | B     | C     | D      | E     | F       | G      |
| species | order | class | phylum | genus | kingdom | family |

Place the letters representing the levels in the correct order in the column below. Some have been done for you.

[2]

|       |
|-------|
| F     |
| ..... |
| C     |
| B     |
| ..... |
| E     |
| ..... |



(c) The classification of a brown bear is shown below. Its scientific name is *Ursus arctos*.

|         |               |
|---------|---------------|
| Kingdom | Animalia      |
|         | Chordata      |
| Class   | Mammalia      |
| Order   | Carnivora     |
|         | Ursidae       |
| Genus   | <i>Ursus</i>  |
|         | <i>arctos</i> |



A wolf has the scientific name *Canis lupus*. Complete the classification table for a wolf. [3]

|         |              |
|---------|--------------|
| Kingdom | Animalia     |
|         | .....        |
| Class   | .....        |
| Order   | Carnivora    |
|         | Canidae      |
| Genus   | .....        |
|         | <i>lupus</i> |

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2. (a) Label the carbon cycle diagram below. Use words from the box. [3]

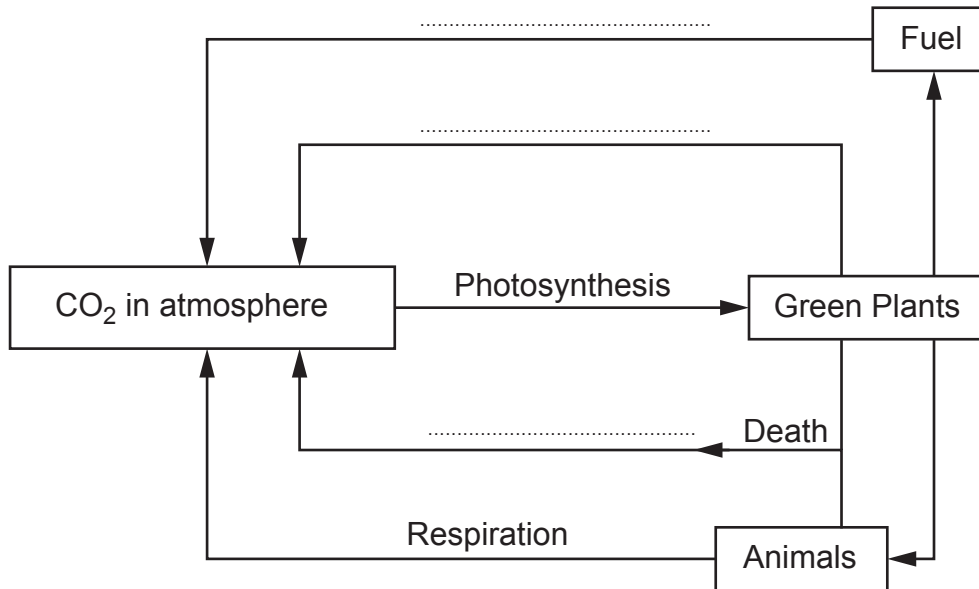
dissolving

evaporating

decay

respiration

combustion



- (b) Complete the following sentences by underlining the correct word in the brackets. [2]

- (i) When plant and animal waste decomposes, ( **sulfates** / **nitrates** / **oxides** ) are added to the soil.
- (ii) Plants use ( **sulfates** / **nitrates** / **oxides** ) in the soil to grow.



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3. Indicator species may be used in an investigation to monitor water pollution at three different locations along a stream.

- (a) The method for an investigation to monitor water pollution in a stream is described below. However, some of the stages are missing.

Complete the method by writing in the missing steps.

[2]

1. Take a tray and a net to the first location.
2. ....
3. Study the organisms in the tray.
4. ....
5. Pour the invertebrates and the water gently back into the stream.
6. Repeat steps 1–5 at the second and third locations.





(b) The following chart is available to you.

**Pollution level**

**Indicator species**

**A** Clean water



Stonefly nymph



Mayfly nymph

**B** Some pollution



Freshwater shrimp



Caddis fly larva

**C** Moderate pollution



Water louse



Bloodworm

**D** High pollution



Sludgeworm



Rat-tailed maggot

**E** Very High pollution – no life

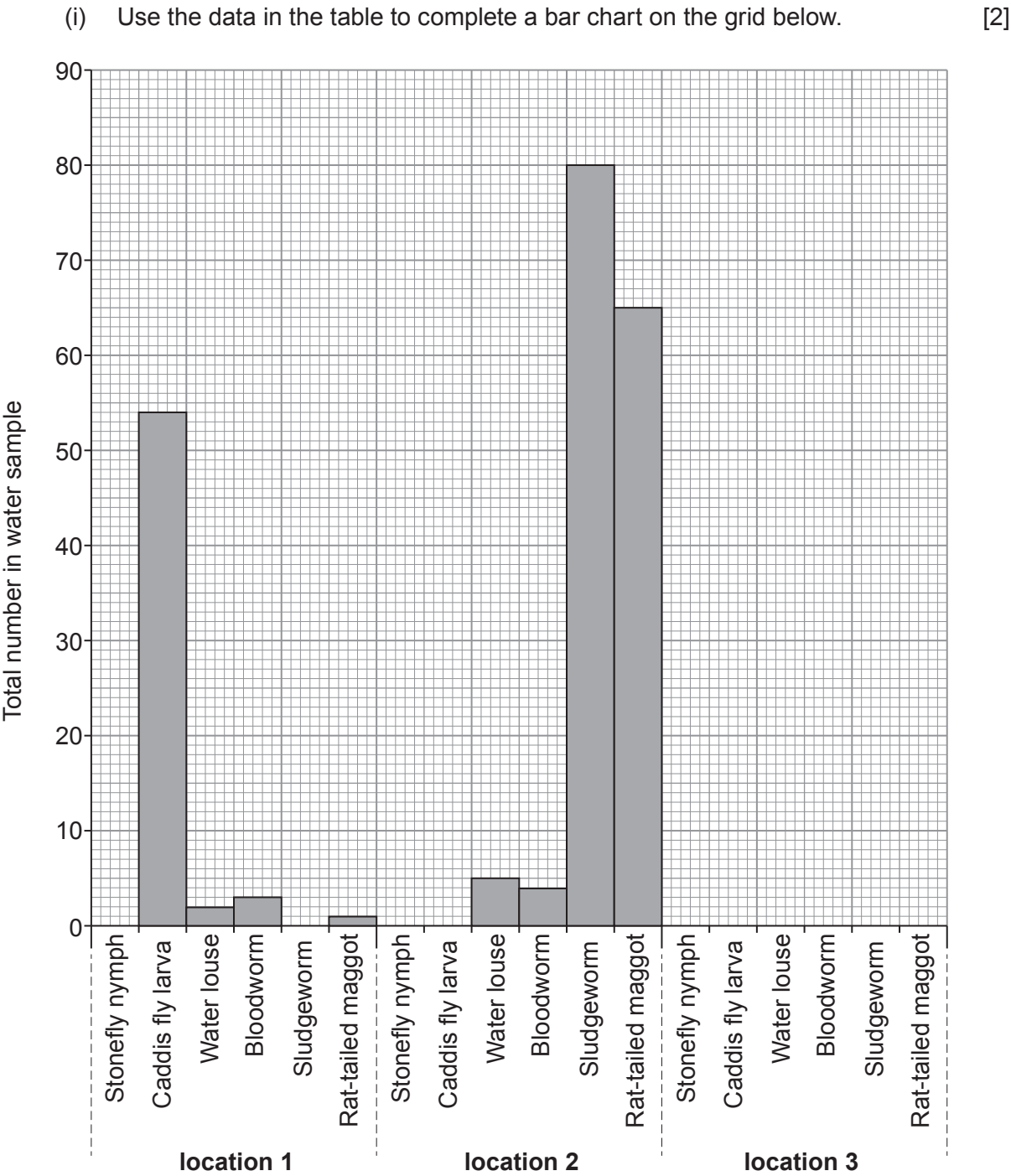


Results from the water sampling at each location are given in the table below.

| Species           | Total number in water sample |            |            |
|-------------------|------------------------------|------------|------------|
|                   | location 1                   | location 2 | location 3 |
| Stonefly nymph    | 0                            | 0          | 31         |
| Caddis fly larva  | 54                           | 0          | 4          |
| Water louse       | 2                            | 5          | 3          |
| Bloodworm         | 3                            | 4          | 2          |
| Sludgeworm        | 0                            | 80         | 1          |
| Rat-tailed maggot | 1                            | 65         | 1          |



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(ii) State which location is the cleanest. [1]

.....

(iii) Give **one** reason for your answer. [1]

.....  
.....  
.....



4. (a) Theories about the Universe have changed over time.

Tick (✓) the boxes next to the names of **two** of these theories.

[2]

Liquid State

☐

Steady State

☐

Loud Bang

☐

Big Bang

☐

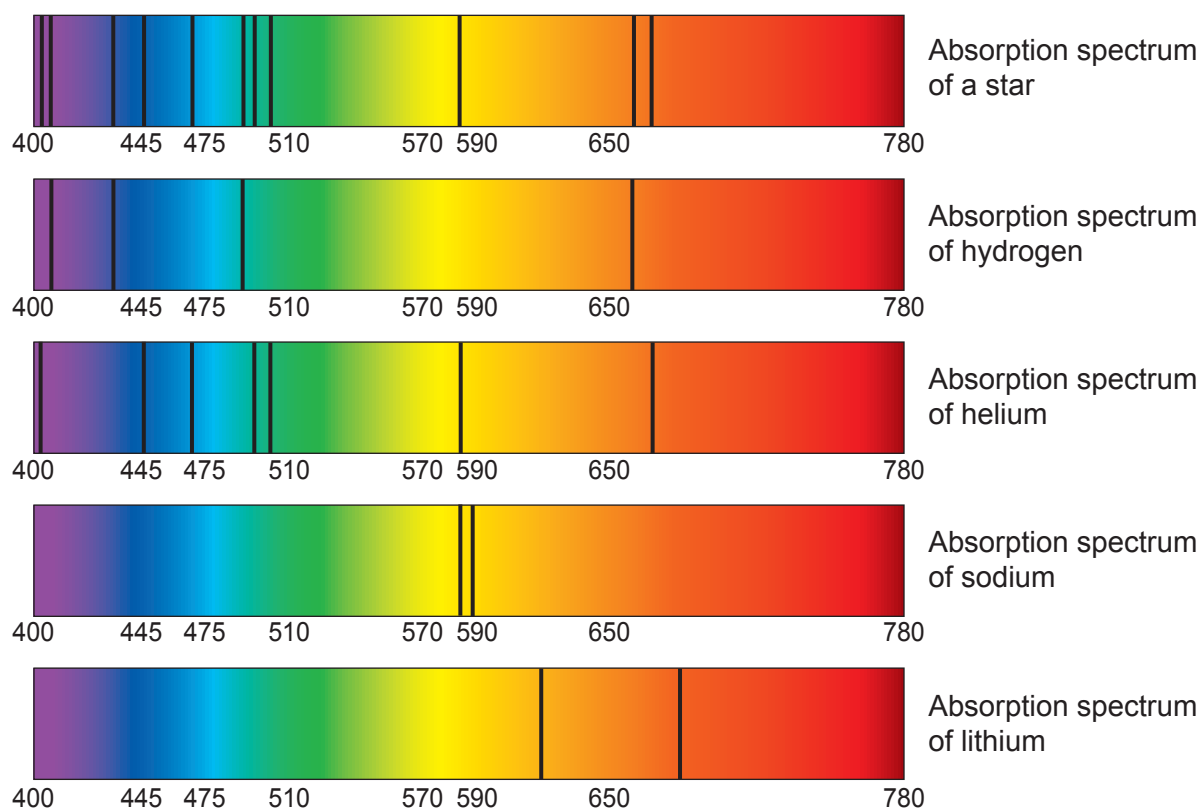
- (b) Images of the Universe are taken using electromagnetic waves.

Name **two** types of electromagnetic wave.

[2]

..... and .....

- (c) The diagram shows a spectrum from a star and four elements.



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Complete the table by placing a tick (✓) if the element is present in the star. Hydrogen has been done for you.

[1]

| Element  | Element is present (✓) |
|----------|------------------------|
| hydrogen | ✓                      |
| helium   |                        |
| sodium   |                        |
| lithium  |                        |

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(d) The table gives information about the planets in our solar system.

|                                         | Mercury | Venus | Earth | Mars | Jupiter | Saturn | Uranus | Neptune |
|-----------------------------------------|---------|-------|-------|------|---------|--------|--------|---------|
| density ( $\text{kg/m}^3$ )             | 5427    | 5243  | 5514  | 3933 | 1326    | 687    | 1271   | 1638    |
| mean distance from the Sun (million km) | 58      | 108   | 150   | 228  | 779     | 1434   | 2873   | 4495    |
| mean speed of orbit (km/s)              | 47.4    | 35.0  | 29.8  | 24.1 | 13.1    | 9.7    | 6.8    | 5.4     |
| mean temperature ( $^{\circ}\text{C}$ ) | 167     | 464   | 15    | -65  | -110    | -140   | -195   | -200    |
| number of moons                         | 0       | 0     | 1     | 2    | 79      | 82     | 27     | 14      |

Use the information in the table to answer the questions below.

(i) Name the coldest **rocky** planet. .... [1]

(ii) State how many **rocky** planets have moons. .... [1]

(iii) Name the gas giant that orbits the Sun with the greatest **speed**. .... [1]

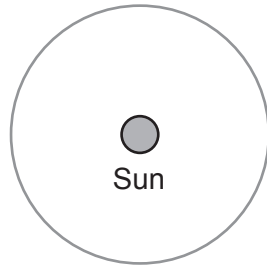
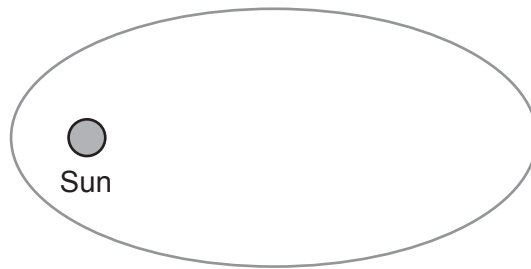
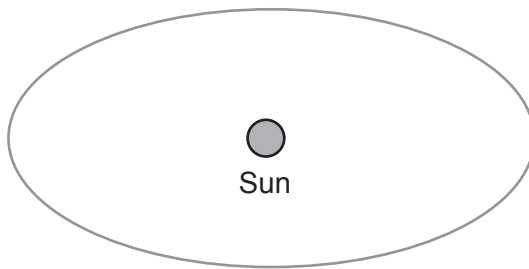
(iv) Eris is a dwarf planet. Its orbit is 10 180 million km away from the Sun.  
Shaun says it is found in the asteroid belt between the orbits of Mars and Jupiter.  
Explain whether you agree with Shaun. [1]

.....  
.....  
.....



(e) Tick (✓) the box next to the shape of the orbit of a comet around the Sun.

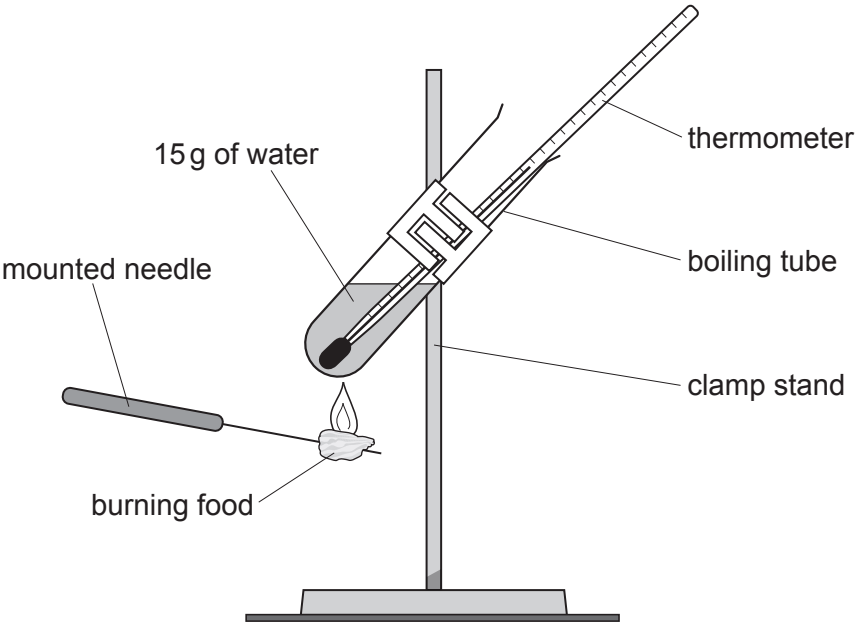
[1]

☐☐☐

10



5. Some students investigated the energy content of food.



Their results are shown in the table below.

| Food          | Mass of food burnt (g) | Initial water temperature (°C) | Final water temperature (°C) | Temperature increase (°C) |
|---------------|------------------------|--------------------------------|------------------------------|---------------------------|
| bread         | 2.5                    | 18                             | 23                           | 5                         |
| cheese        | 4.0                    | 17                             | 31                           | 14                        |
| cornflakes    | 3.0                    | 17                             | 29                           | 12                        |
| oatie biscuit | 5.0                    | 18                             | 25                           | 7                         |

(a) Use the equation

energy released = 15 × 4.2 × temperature increase

to calculate the energy released by the bread.

[2]

energy released = ..... J





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- (b) The students were asked to use their results to calculate the energy provided by a snack consisting of 15 g of oatie biscuits and 20 g of cheese. Complete the tables below to work out their answer.

[3]

| Energy from <b>5 g</b> of oatie biscuits (J) | Energy from <b>15 g</b> of oatie biscuits (J) |
|----------------------------------------------|-----------------------------------------------|
| 441                                          | .....                                         |

| Energy from <b>4 g</b> of cheese (J) | Energy from <b>20 g</b> of cheese (J) |
|--------------------------------------|---------------------------------------|
| 882                                  | .....                                 |

Space for working

total energy from snack = ..... J

|   |
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| 5 |



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6. All drugs can have side effects. New drugs undergo testing before they can be prescribed by doctors.

The following terms are used in drug development studies.

Join each term to its description with a straight line.

[3]

| Term               | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| Blind study        | an inactive substance made to look like the real drug                           |
| Double-blind study | when the patient and the researcher do not know if they are using the real drug |
| Placebo            | when only the researcher does not know if they are using the real drug          |
|                    | when the patient does not know if they are given the real drug                  |

3



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7. Some of the methods used in the treatment and diagnosis of disease are included in the table below.

Place **one** tick (✓) in each row underneath the method used by each treatment or diagnosis.

[4]

| Treatment or diagnosis | Uses high intensity light waves | Uses short wavelength electromagnetic waves | Uses high frequency sound waves | Uses strong magnetic fields | Uses drugs |
|------------------------|---------------------------------|---------------------------------------------|---------------------------------|-----------------------------|------------|
| chemotherapy           |                                 |                                             |                                 |                             |            |
| ultrasound             |                                 |                                             |                                 |                             |            |
| MRI scan               |                                 |                                             |                                 |                             |            |
| CAT scan               |                                 |                                             |                                 |                             |            |

|   |
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|   |
| 4 |



8. Vaccines are used to protect against disease.

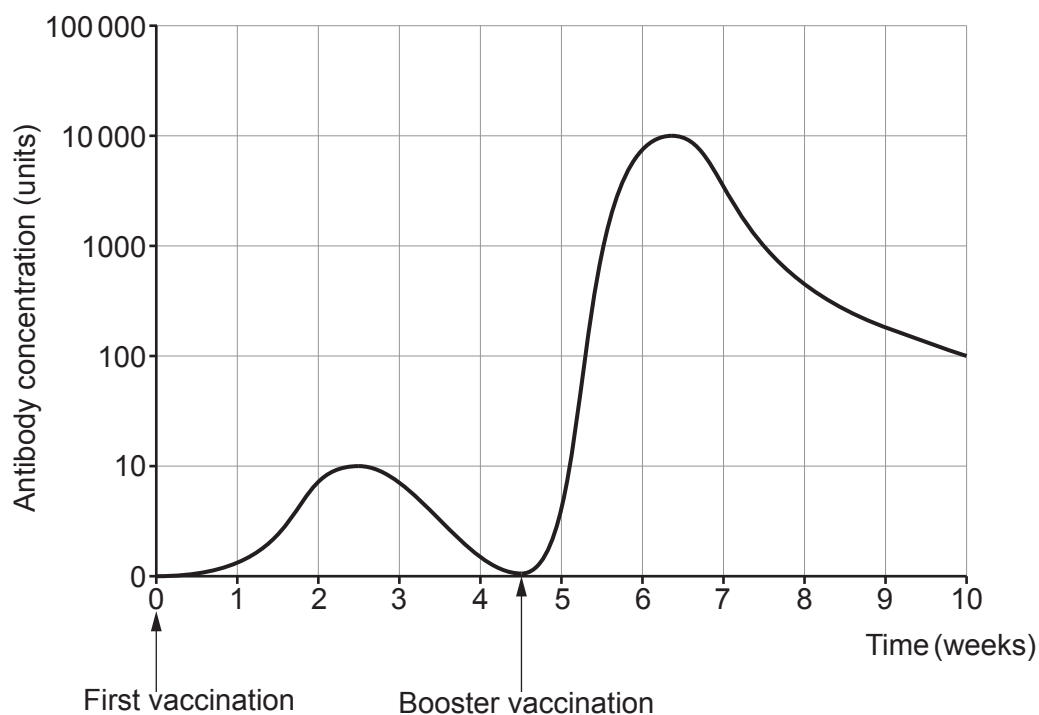
(a) Complete the following sentences about how the body reacts after infection or vaccination by underlining the correct term in the brackets.

(i) Vaccines contain molecules called ( **antibodies** / **antigens** / **memory cells** ). [1]

(ii) White blood cells respond to these molecules and produce ( **antibodies** / **antigens** / **pathogens** ). [1]

(iii) ( **Antibiotics** / **antigens** / **memory cells** ) are produced following infection or a vaccination. [1]

(b) The graph below shows the immune response over a 10-week period when a vaccine was given as two doses.



Describe the changes in antibody concentration over 10 weeks using data from the graph. [6 QER]

.....

.....

.....

.....



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|   |
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|   |
| 9 |



Section B

Refer to the separate Resource Folder to answer questions 9 and 10.

9. Jake’s heart rate, breathing rate and airflow were monitored during a training session. Use the information in **Table 2** to answer the following questions.
- (a) State the value of the resting heart rate. [1]  
heart rate = ..... bpm
- (b) State the time at which the exercise stopped. [1]  
time = ..... min
- (c) Calculate the time the exercise lasted. [1]  
time = ..... min
- (d) (i) Calculate the drop in heart rate between 14 and 15 minutes. [1]  
drop in heart rate = ..... bpm
- (ii) Assuming that the heart rate drops at the same rate, calculate the heart rate after 16 minutes. [1]  
heart rate after 16 minutes = ..... bpm
- (e) Estimate the breathing rate after 16 minutes. [1]  
breathing rate = ..... breaths per minute

|   |
|---|
|   |
| 6 |



10. (a) Dave says the distances of each leg in a long-distance triathlon are all double those in a middle-distance triathlon.

Tom says the distances of each leg in an Olympic triathlon are double those in a sprint triathlon.

Use data from **Table 1** to explain whether Dave or Tom is correct.

[2]

.....

.....

.....

.....

- (b) Piera says Jake breathed in less air per minute at 4 minutes than at 3 minutes because his breathing rate was lower.

Petula disagrees because Jake's airflow per breath was greater at 4 minutes than at 3 minutes.

Use data from **Table 2**, and the equation below it, to explain whether you agree with Piera or Petula.

[2]

.....

.....

- (c) Use the information in **Table 4** and equations from pages 4 and 5 to answer the following questions.

- (i) Calculate Malcolm's BMI.

[2]

BMI = .....

- (ii) Calculate Malcolm's maximum heart rate.

[1]

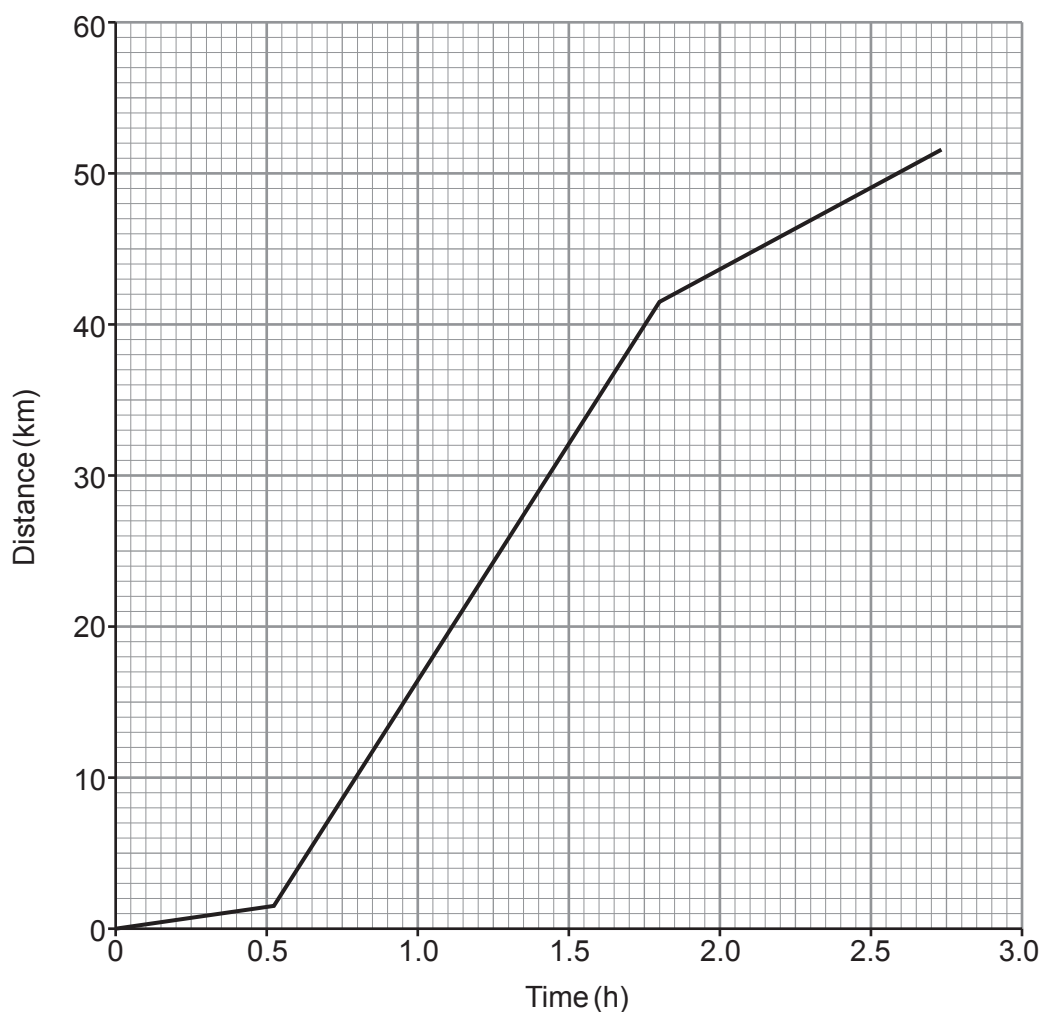
maximum heart rate = ..... bpm



- (d) Karen wants to exercise in the aerobic zone. Use the information in **Tables 3 and 4** and on page 4 to determine the range in heart rate she should aim for. [1]

heart rate range is from ..... bpm to ..... bpm.

- (e) Data in the **Triathlon times** section (page 5) and **Table 1** was used to plot distance-time graphs for each of the triathlons described. There were three sections in each graph to show the swim, cycle and run legs. The graphs did not include transition times. One of the graphs is shown below.





Determine which type of triathlon is represented by the graph and give **two** reasons for your answer. [3]

Type of triathlon: .....

Reason 1: .....

Reason 2: .....

- (f) Use the information about Triathlon times on page 5, the information in **Table 1** and an equation on page 7 to calculate the mean speed during an Ironman triathlon. [3]

mean speed = ..... km/h

- (g) When a triathlete mounts their bike they are travelling at 1.5 m/s. Use the information in **Table 5** and an equation on page 7 to calculate the acceleration along level ground in a cycle leg. [3]

acceleration = ..... m/s<sup>2</sup>

- (h) It is thought that the higher the age group of the triathlete, the lower the mean speed in the cycle leg. Use data from **Graph 1** to explain whether this statement is true for all age groups. [2]

.....

.....

.....

.....

**END OF PAPER**





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|   | Group | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 0 |
|---|-------|---|---|---|---|---|---|---|---|
| 1 | 1     |   |   |   |   |   |   |   |   |
| 2 | 2     |   |   |   |   |   |   |   |   |
| 3 | 3     |   |   |   |   |   |   |   |   |
| 4 | 4     |   |   |   |   |   |   |   |   |
| 5 | 5     |   |   |   |   |   |   |   |   |
| 6 | 6     |   |   |   |   |   |   |   |   |
| 7 | 7     |   |   |   |   |   |   |   |   |
| 0 | 0     |   |   |   |   |   |   |   |   |

|                                                                                                     |           |     |           |     |           |     |           |     |          |     |            |     |            |     |            |     |           |      |          |      |         |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
|-----------------------------------------------------------------------------------------------------|-----------|-----|-----------|-----|-----------|-----|-----------|-----|----------|-----|------------|-----|------------|-----|------------|-----|-----------|------|----------|------|---------|-----|--------|-----|---------|-----|-----------|-----|-----------|-----|-----------|-----|---------|-----|---------|
| <div><div><div><div><div>1</div><div>H</div><div>Hydrogen</div><div>1</div></div></div></div></div> |           |     |           |     |           |     |           |     |          |     |            |     |            |     |            |     |           |      |          |      |         |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
| 7                                                                                                   | Li        | 9   | Be        |     |           |     |           |     |          | 11  | B          | 12  | C          | 14  | N          | 16  | O         | 19   | F        | 20   | Ne      |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
|                                                                                                     | Lithium   |     | Beryllium |     |           |     |           |     |          |     | Boron      |     | Carbon     |     | Nitrogen   |     | Oxygen    |      | Fluorine |      | Neon    |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
| 23                                                                                                  | Na        | 24  | Mg        |     |           |     |           |     |          | 27  | Al         | 28  | Si         | 31  | P          | 32  | S         | 35.5 | Cl       | 40   | Ar      |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
|                                                                                                     | Sodium    |     | Magnesium |     |           |     |           |     |          |     | Aluminium  |     | Silicon    |     | Phosphorus |     | Sulfur    |      | Chlorine |      | Argon   |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
| 39                                                                                                  | K         | 40  | Ca        | 45  | Sc        | 48  | Ti        | 51  | V        | 52  | Cr         | 55  | Mn         | 56  | Fe         | 59  | Co        | 59   | Ni       | 63.5 | Zn      | 65  | Ga     | 70  | Ge      | 73  | As        | 75  | Se        | 79  | Selenium  | 80  | Br      | 84  | Kr      |
|                                                                                                     | Potassium |     | Calcium   |     | Scandium  |     | Titanium  |     | Vanadium |     | Chromium   |     | Manganese  |     | Iron       |     | Cobalt    |      | Nickel   |      | Copper  |     | Zinc   |     | Gallium |     | Germanium |     | Arsenic   |     | Selenium  |     | Bromine |     | Krypton |
| 86                                                                                                  | Rb        | 88  | Sr        | 89  | Y         | 91  | Zr        | 93  | Nb       | 96  | Mo         | 99  | Tc         | 101 | Ru         | 106 | Pd        | 108  | Ag       | 112  | Cd      | 115 | In     | 119 | Sn      | 122 | Sb        | 127 | Te        | 128 | Tellurium | 127 | I       | 131 | Xe      |
|                                                                                                     | Rubidium  |     | Strontium |     | Yttrium   |     | Zirconium |     | Niobium  |     | Molybdenum |     | Technetium |     | Ruthenium  |     | Palladium |      | Silver   |      | Cadmium |     | Indium |     | Tin     |     | Antimony  |     | Tellurium |     | Iodine    |     | Xenon   |     |         |
| 133                                                                                                 | Cs        | 137 | Ba        | 139 | La        | 179 | Hf        | 181 | Ta       | 184 | W          | 186 | Re         | 190 | Os         | 195 | Pt        | 197  | Au       | 201  | Hg      | 204 | Pb     | 207 | Pb      | 209 | Bi        | 210 | Po        | 210 | At        | 222 | Rn      |     |         |
|                                                                                                     | Caesium   |     | Barium    |     | Lanthanum |     | Hafnium   |     | Tantalum |     | Tungsten   |     | Rhenium    |     | Osmium     |     | Iridium   |      | Gold     |      | Mercury |     | Lead   |     | Lead    |     | Bismuth   |     | Polonium  |     | Astatine  |     | Radon   |     |         |
| 223                                                                                                 | Fr        | 226 | Ra        | 227 | Ac        |     |           |     |          |     |            |     |            |     |            |     |           |      |          |      |         |     |        |     |         |     |           |     |           |     |           |     |         |     |         |
|                                                                                                     | Francium  |     | Radium    |     | Actinium  |     |           |     |          |     |            |     |            |     |            |     |           |      |          |      |         |     |        |     |         |     |           |     |           |     |           |     |         |     |         |

Key

## Key

- relative atomic mass

| Ar | Symbol | Name | atomic number |
|----|--------|------|---------------|
| 7  |        |      |               |